

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Provisioned and managed cloud infrastructure with Terraform; owned CI/CD pipelines via GitHub Actions across 7 production deployments in TypeScript/Node.js with Docker and Sentry observability. - Engineered multi-model LLM pipelines (Claude, GPT-4o, Gemini) via MCP; built document processing (PDF gen, email parsing, S3) and full Playwright automated test coverage.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling 100+ researchers to explore 500GB datasets; managed GCP infrastructure and deployment pipelines. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Automated deployments on AWS and GCP, integrated Grafana monitoring dashboards, and led model red-teaming to strengthen system reliability. - Built 5+ AI-driven workflows including health claim automation using OCR, NLP, and ML; improved processing accuracy by 25%.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built distributed Kafka + Spark + Hadoop pipelines for batch and real-time ingestion, cutting ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

PulseRank	Real-Time Leaderboard	2026
- Built event-driven platform in TypeScript/NestJS/Next.js: MySQL outbox, MongoDB TTL projections, Redis sorted sets; Testcontainers integration tests for multi-store consistency. - Shipped OpenTelemetry + Prometheus observability on all hot paths; load-tested with k6; deployed via GitHub Actions CI/CD.		
Demand Paging – Xinu OS	Systems Programming	Oct 2024 – Dec 2024
- Engineered 8-store virtual memory system in C for Xinu OS with Second-Chance page replacement; reduced page faults by 25%. - Added 7+ kernel system calls, increasing memory performance by 15% using inverted page tables and a custom fault handler.		
VectorLift	Semantic Search Engine	2026
- Built semantic search on MS MARCO in Python: BM25 (Elasticsearch), dense bi-encoder with in-batch negatives, cross-encoder re-ranker. - Served via FastAPI with per-stage latency tracking; Kafka-driven corpus indexing into Qdrant; Docker Compose for local infra.		
LLM Eval Harness	Independent Project	2025
- Built vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini across 2,000 examples and 3 task types in Python. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		

Languages and Technologies

- Languages:** Python, C, TypeScript, Java, JavaScript, Bash, SQL
- Cloud & Infra:** GCP, AWS, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Grafana, OpenTelemetry, Prometheus
- Testing & Reliability:** Testcontainers, Playwright, k6, unit/integration testing, red-teaming, TDD
- Backend & Data:** Node.js, NestJS, FastAPI, Spring Boot, PostgreSQL, MySQL, Redis, Kafka, Spark

Additional Experience and Awards

- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.